A 79-variable quartic under rational *abc*

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Abstract

Under rational *abc*, the specified elliptic-group and prime-selection premises, and the simultaneous-twist theorem of Jaswal, we construct a quartic family in 79 total rational variables defining the integers for every sufficiently large fixed odd coefficient shift. It is a sum of 71 primitive quadratic squares with 1139 ordinary monomials. A complete five-field reference chart removes two coordinates from the reciprocal 81-variable construction; the new pole equations reduce their fixed coefficients without changing their count. This edition uses the corrected reference coefficient table, including its previously omitted fS term. All exceptional strata, half-integral indices and pole realization conditions are included. The defining shift is ineffective.

1 Statement and exact premises

All variable counts include the input. Heights of integer polynomials are maximum absolute coefficients after coefficient parameters have been substituted. Put

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d = 3 - \rho, \quad D = d^2 = 11 - 6\rho, \qquad E : y^2 = x^3 - Dx,$$

$$R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0 = (a, b) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32}\right).$$

The retained arithmetic premises have the following scope.

Elliptic group. We assume $2E(K) = \mathbb{Z}P$ and P non-torsion. Hence every point has an index in $\frac{1}{2}\mathbb{Z}$, as proved in [4, Corollary 1.3]. The reciprocal pair below is not restricted to $2E(K)$.

Pole realization from simultaneous twists. We use Jaswal's Theorem 1.2 in Hexagon version 2 [3]: for coprime positive squarefree $\beta_0 > 1$, $\theta \geq 1$, both 1 (mod 8) and each a square modulo every prime of the other, there is $r > 0$ such that $y^2 = x^3 - r^2x$ and $y^2 = x^3 - r^2\theta^2 x$ both have an infinite-order rational point with abscissa in $\beta_0\mathbb{Q}^{\times 2}$. We do not use that paper's other height or prime-pattern assertions. The precise application, including the square input case $\beta_0 = 1$, is proved in [2]. It supplies a witness for

$$e^2 + b_0p^2 = 1, \qquad uv = e(1 - b_0), \qquad u^2 - v^2 = b_0q^2$$

with $upq \neq 0$ whenever

$$0 < b_0 < 1, \quad b_0 \text{ has square reduced denominator}, \quad b_0 \in \mathbb{Q}_2^{\times 2}, \quad v_2(b_0) \geq 3. \tag{1}$$

The same equations forbid a negative odd valuation at every odd prime, by the elementary proof in [2]. Thus the older restricted-converse pole input is not required for this family. The stated simultaneous-twist input and the inherited prime and group premises remain explicit; this paper does not declare their complete external audit finished.

Prime selection. We retain the fixed-data simultaneous-prime statement of [8, 9]. It applies to any fixed finite list of distinct rational contact coefficients after a fixed square clearing of denominators. It allows prescribed divisibility of the square denominator, finite avoidance, an arbitrarily large fixed lower bound on the rational argument, and the Gaussian congruences and conjugate annihilating orders needed for the common index. In the source normalization these orders U_*, V_* and preparation modulus L_* have

$$\gcd(U_*, V_*) = 1, \quad \gcd(U_*, L_*) = 16, \quad \gcd(V_*, L_*) = 1,$$

so $h \equiv 0 \pmod{U_*}$, $h \equiv 1 \pmod{V_*}$ and $h \equiv h_* \pmod{L_*}$ are compatible when $64 \mid h_*$. The source also retains the prescribed compact real neighborhoods in that progression. All finite data and rational coefficients are fixed before prime selection. We check the new coefficient list and all subsequent uses in Section 5.

Additional hypothesis. Ordinary rational *abc* is assumed. The eight-contact estimate below is not asserted to follow from the separate weak-*abc*/Szpiro hypothesis of the thirteen-variable family. No modular contact-height premise is needed here.

Theorem 1.1. *Under these premises there is a computable family $P_B \in \mathbb{Z}[X, Y_1, \dots, Y_{78}]$, indexed by odd integers $B \geq 3$, such that every member has degree four, content one, 1139 nonzero coordinate monomials, and exact height $\max(C_*, B^2 + C_0)$, where*

$$C_* = 116167112798432587389449032604886634366655869137034483200,$$
$$C_0 = 17471454977902235928100604664093317385559865261753175317559.$$

There is a fixed B_* such that for every odd $B \geq B_*$,

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{78} : P_B(x, \boldsymbol{y}) = 0.$$

The coefficient parameter B is not a quantified variable. No numerical sufficient B_* is given; the $B = 3$ table is an algebraic fixture, not a certified defining member. A separate 81-variable presentation using the older reference chart and the same new poles has height $\max(4153456, B^2 + 630511)$ and the same input projection for every fixed B.

2 The quadratic system

A field variable denotes two rational coordinates in the basis $1, \rho$. A bar denotes conjugation. The notation S^2 denotes an ordinary product, whereas the variables named F_H, G_H, F_U, F_V are stored field values. Every displayed equation has degree at most two in the retained rational coordinates.

2.1 The common reference point

Retain $f_H, S, T_H, F_H, G_H \in K$. Use the four exact field quadrics of [1, Theorem 1.1 and corrected coefficient table], with substitutions $(f, S, B, w, T) = (f_H, S, F_H, G_H, T_H)$:

$$F_H = f_H^2, \qquad Q_1 = 0, \qquad MG_H + L = 0, \qquad Q_3 = 0.$$

Every coefficient is fixed by that table and the accompanying `reference-equations.json`. In particular L includes

$$\left(\frac{171357}{235298} + \frac{2519877}{7529536}\rho\right) f_H S.$$

The complete proof gives $H, H-P$ finite with nonzero abscissas and

$$f_H = 1/x(H), \quad S = y(H)/x(H), \quad T_H = y(H-P)/x(H-P),$$

$$F_H = 1/x(H)^2, \qquad G_H = 1/x(H-P)^2.$$

It excludes the vanishing coefficient $M = 0$ over K by an exact quintic factorization and a nonsquare discriminant norm, not by a new inverse variable or a generic-open-set restriction. Conversely every allowed reference point supplies these five coordinates. Write $g_H = 1/x(H-P)$ only for interpretation; it is not retained. No global nonvanishing of S, T_H is assumed.

2.2 Two points and their difference

Retain $f, h, g, k, s \in K$, with stored squares F_U, F_V, and impose

$$F_U = f^2, \quad F_V = g^2, \quad h^2 = f - DfF_U, \quad s(f-g) = h+k, \quad s(h-k) = 1 - D(F_U + fg + F_V). \tag{2}$$

The complete algebraic proof in [4, Proposition 2.1] identifies points $U, V \in E(K)$, including the identity, with $U \neq V$. Their finite difference $A = U - V$ has

$$c = x(A) = s^2 + D(f+g), \qquad y(A) = Dh - s(Df + c).$$

Neither U, V nor A is assumed to have square abscissa. The theorem includes the tangent case $f = g$, and excludes $(f, h) = (g, k) = (0, 0)$. Define one retained rational coordinate

$$T = 2\,\mathrm{Re}(s^2 + D(f+g)) = \mathrm{Tr}\, x(A). \tag{3}$$

This is a quadratic equation. The index n of A belongs to $\frac{1}{2}\mathbb{Z}$, and may be zero.

2.3 Ratios, comparisons and the additive scale

Retain $L, L_0 \in K$ with

$$hL + fS = 0, \qquad (1 - DF_U)L + hS = 0, \tag{4}$$

$$kL_0 + gT_H = 0, \qquad (1 - DF_V)L_0 + kT_H = 0. \tag{5}$$

At formal nonidentity points these are the usual ratios, and at an identity numerator they force the ratio to be zero. No division by h or k is made on torsion components.

Retain the input $X \in \mathbb{Q}$, a comparison coordinate $I_0 \in \mathbb{Q}$ and a slope $\lambda \in \mathbb{Q}$. Write $L = L_r + \rho L_i$ and impose

$$I_0 - L_r + X^2 + B = 0, \qquad \beta = L_r - I_0 = X^2 + B, \qquad I = I_0 + \rho L_i, \qquad C = L - \overline{L_0}.$$

Retain $C_2, I_2 \in K$ in addition to the already stored F_H, G_H, and two scalar height coordinates u_h, v_h, with

$$C_2 = C^2, \quad I_2 = I^2, \qquad u_h = \beta^2 + T^2, \quad v_h = \beta T.$$

The scale $a_1 = \beta^4 + T^4 = u_h^2 - 2v_h^2$ is not retained as a coordinate. Use arguments

$$t_0 = \lambda, \qquad t_1 - \lambda = u_h^2 - 2v_h^2. \tag{6}$$

Only t_1 is a further coordinate. Notice that $a_1 \geq B^4 \geq 81$.

2.4 Contact and pole equations

For each argument $t = t_j$ retain scalars R, Z, w, w_2, w_3, field values $A_1, A_2, \Omega = \omega_0 + \rho\omega_1$, and scalars $\eta, b_0, e, u, p, v, q, p_2, q_2$. Impose

$$R = t^2 - 2t, \qquad RZ + Z - R^2 + 35R = 0, \qquad (R+1)w = Z^2 + 24Z, \tag{7}$$

$$w_2 = w^2, \qquad w_3 = ww_2, \tag{8}$$

$$w_3A_1 = w_2^2 + w_3C_2 + w_2F_H + F_U^2, \tag{9}$$

$$w_3A_2 = w_2^2 + w_3I_2 + w_2\overline{G_H} + \overline{F_V}^2, \tag{10}$$

$$(R+1)\Omega = A_1A_2, \tag{11}$$

$$\omega_0^2 - 2\omega_0\eta - 2\omega_1^2 = 0, \qquad w_3b_0 = \eta^2, \tag{12}$$

$$p_2 = p^2, \qquad e^2 + b_0p_2 = 1, \tag{13}$$

$$uv = e(1 - b_0), \qquad q_2 = q^2, \tag{14}$$

$$u^2 - v^2 = b_0q_2. \tag{15}$$

In each factor equation a scalar is interpreted as a field element. The two blocks have separate pole coordinates. The last five equations are the quadratic presentation of [2]; all zero auxiliary branches are included in its local obstruction proof. Realization of their actual prepared inputs is established below, not by identifying their full zero set with that of the earlier pole equations.

3 Domains and local extraction

Put $r = R + 1$, $z = Z + 37$. The contact equations give

$$r = (t-1)^2, \quad r(z-r) = 36, \quad rw = z^2 - 50z + 481.$$

Thus $r \neq 0$. With

$$P_4(r) = (r-1)(r-4)(r-9)(r-36), \qquad \Theta = \{-5, -2, -1, 0, 2, 3, 4, 7\},$$

one has $P_4(r)/r^2 = z^2 - 50z + 481$ and $F(t) := \prod_{\theta\in\Theta}(t-\theta) = wr^3$. If $w = 0$, the factor equations give $F_U^2 = \overline{F_V}^2 = 0$, hence $f = g = 0$; the cubic gives $h = 0$, the secant gives $k = 0$, and the last equation of (2) gives $0 = 1$. Therefore $rwF(t) \neq 0$ at every rational solution, and $t \notin \Theta$.

For interpretation only, define

$$q_* = w^{-1} = \frac{(t-1)^6}{F(t)}, \quad R_* = \frac{F(t)}{(t-1)^{10}}, \quad T_i = A_i/w, \quad W = T_1T_2 = \frac{r}{w^2}\Omega, \quad \tau = \frac{r}{w^2}\eta.$$

Then

$$T_1 = 1 + q_*C^2 + q_*^2f_H^2 + q_*^4f^4,$$
$$T_2 = 1 + q_*I^2 + q_*^2\overline{gH}^2 + q_*^4\overline{g}^4, \qquad b_0 = R_*\tau^2.$$

The cleared trace equation is preserved under this common rescaling. None of W, η, b_0 is assumed globally nonzero at this stage.

For $t \notin \Theta$, let $M(t)$ be the product of primes $p > 76176$ with positive odd valuation at one of $t - \theta$. At such a prime, all contact differences and $t - 1$ are units. Write $\nu = v_p(F(t)) > 0$, odd; then $v_p(q_*) = -\nu$. Every nonzero reciprocal abscissa has even valuation at these good places. This follows directly from $h^2 = f - Df^3$ if $v(f) \neq 0$; valuation zero is even too.

The same statement holds for the common reference point. A zero reciprocal contributes no nonconstant term.

The finite nonconstant valuations in each factor are respectively odd, 2 modulo 4, and 4 modulo 8, and differ from the constant term of valuation zero. Thus both factors are nonzero and have valuation at most zero at every place above p. Let $\ell = \max_{\mathfrak{p}|p} v_{\mathfrak{p}}(W) \le 0$. The trace equation forces the rational component of W and τ to be nonzero, and

$$\tau = \operatorname{Tr}(W^{-1})^{-1}, \qquad v_p(b_0) = \nu - 2v_p \operatorname{Tr}(W^{-1}) \le \nu + 2\ell.$$

This valuation is odd. The pole obstruction forces it positive, so $\ell > -\nu/2$. At a place attaining ℓ, each nonconstant summand valuation exceeds $-\nu/2$. Consequently

$$v(C), v(I) > \nu/4, \quad v(f_H), v(\overline{g_H}) > 3\nu/4, \quad v(f), v(\overline{g}) > 7\nu/8.$$

The reciprocal valuations are positive even integers, and C, I have order at least one. In particular the four indicated points are in the appropriate primary or conjugate formal groups, with the identity convention at zero reciprocals.

Write the indices of H, U, V, A as $h_*, u_*, u_* - n, n$ in $\frac{1}{2}\mathbb{Z}$. At a consumed prime, $H, H - P$ are nonidentity formal points and therefore non-torsion. Multiplying the integer-index group identities by two kills every possible two-torsion component. The full formal-logarithm ratios are

$$u_*/h_*, \qquad (u_* - n)/(h_* - 1).$$

At least one denominator is a p-adic unit, and the numerator indices are p-integral. Thus one full ratio is integral. The order-four relative logarithm estimate in [4] first makes its plain ratio integral. The test $C \equiv 0 \pmod p$ then gives integrality of the other plain ratio, hence of its full ratio. Only after this step are the relative errors used as absolute congruences. Subtracting gives

$$n = h_* \frac{u_*}{h_*} - (h_* - 1)\frac{u_* - n}{h_* - 1} \equiv L \equiv \beta \pmod p.$$

This proves that, if $\beta \neq n$ and N is the positive reduced numerator of $\beta - n$, then

$$M(t_0) \mid N, \qquad M(t_1) \mid N. \tag{16}$$

Inert primes are included: if the required formal groups are incompatible, no solution occurs at that prime. No assertion that the two radicals are coprime is needed.

4 Heights and the uniform sufficient shift

Lemma 4.1. *Under rational abc, fixed constants C_0, C_1 satisfy*

$$h(t) \le \tfrac{1}{3}\log M(t) + C_0 \quad (t \notin \Theta), \qquad h(1 : u : v) \le \tfrac{4}{3} h(u^4 + v^4) + C_1 \quad (u, v \in \mathbb{Q}).$$

Proof. For $t = a/b$ reduced use $G(a, b) = \prod_{\theta \in \Theta}(a - \theta b)$. Granville's Proposition 1 [5], with exponent loss $1/2$, gives $\operatorname{rad}(G) \gg \max(|a|, b)^{11/2}$. Since $\operatorname{rad}(m)^2 \le |m| \operatorname{sf}(m)$ and $|G| \ll \max(|a|, b)^8$, its odd part is at least a fixed multiple of $\max(|a|, b)^3$. Outside the fixed exceptional primes it is exactly $M(t)$: two primitive contact factors cannot vanish together, and a prime in b divides none. This proves the first assertion, including bounded heights by increasing C_0. The form has eight distinct factors and $G(a, b) \neq 0$, so all source hypotheses are met. The author's corrigendum changes Theorems 8.1 and 8.3, not this proposition; neither corrected theorem is used here.

For the second assertion, the zero-coordinate cases are immediate. Otherwise write $u = gA/D_0$, $v = gB_0/D_0$ with $\gcd(A, B_0) = \gcd(g, D_0) = 1$, $g, D_0 > 0$. Set $H = \max(|A|, |B_0|)$,

$N_0 = A^4 + B_0^4$, and $k = \gcd(N_0, D_0^4)$, $m_0 = N_0/k$, $n_0 = D_0^4/k$. Then $u^4 + v^4 = g^4 m_0/n_0$ is reduced. Put $M = \max(g^4 m_0, n_0)$. Rational *abc* at exponent $16/15$ and $\operatorname{rad}(N_0) \le D_0 m_0$ give $H^{7/4} \le K^{15/16} D_0 m_0$. Also $D_0^4 \le 2n_0 H^4/m_0$, whence $H^{3/4} \le 2^{1/4} K^{15/16} m_0^{3/4} n_0^{1/4}$. Finally $\max(gH, D_0) \le 2^{1/4} H (M/m_0)^{1/4}$. Taking logarithms bounds $h(1 : u : v)$ by a fixed constant plus $\frac{3}{4}\log m_0 + \frac{1}{3}\log n_0 + \frac{1}{4}\log M \le \frac{4}{3}\log M$. This accounts for all common numerator and denominator cancellation. □

Suppose $\beta \neq n$. By (6), (16) and the elementary additive height inequality,

$$h(a_1) \le h(t_0) + h(t_1) + \log 2 \le \tfrac{2}{3}\log N + 2C_0 + \log 2.$$

Writing $H_2 = h(1 : \beta : T)$, the preceding lemma yields

$$H_2 \le \tfrac{8}{9}\log N + C_2.$$

Because $2n \in \mathbb{Z}$, the reduced numerator satisfies $N \le 2(1 + |n|)e^{h(\beta)}$. Hence

$$H_2 \le 8\log(2(1 + |n|)) + 9C_2.$$

The finite point $A = U - V$ satisfies the all-point trace theorem [4, Corollary 1.3], including torsion and half-integral indices. Thus

$$n^2 \le 8A_0 \log(2(1 + |n|)) + 9A_0 C_2 + J_0.$$

This bounds $|n|$ by a fixed constant and then bounds H_2 by a fixed constant, independent of B and every witness. But $h(\beta) \ge \log\beta \ge \log B$, since $\beta = X^2 + B$. One sufficiently large fixed B_* excludes $\beta \neq n$ for all $B \ge B_*$. Therefore $\beta \in \frac{1}{2}\mathbb{Z}$. Writing $X = a/b$ reduced gives $b^2 \mid 2a^2$, hence $b^2 \mid 2$ and $b = 1$. This proves soundness. The replacement of $2X^2 + B$ by $X^2 + B$ is essential for this denominator argument with half-indices.

5 Completeness

Fix an integer X and an odd $B \ge 3$. Set $n = \beta = X^2 + B > 0$ and $A = [n]P$. This fixes $T = \operatorname{Tr} x(A)$ and $a_1 = \beta^4 + T^4 \ge 81$ before selecting primes. Choose a nonexceptional reference multiple $H = [h]P$, with its initial index divisible by 64, in fixed compact real and finite-place neighborhoods. Set $U = [hn]P$ and $V = [(h-1)n]P$. All are in the doubled group, so the usual square-abscissa assertions are available for these particular witnesses, although they were not imposed on all solutions. Avoid finitely many forbidden reference points and denominators. The reciprocal-pair secant or tangent lemma then supplies its coordinates; A is never the identity.

The new fixed list of contact coefficients in the slope variable is

$$\Theta \ \cup \ (\Theta - a_1).$$

The two lists are disjoint since their span is 12 and $a_1 \ge 81$. Choose an integer $D_1 > 0$ whose square clears these rational coefficients, sufficiently divisible at every preparation prime. Apply the retained prime-selection input to this list with $\lambda = u/(D_1^2 w^2)$. Its prime numerators are exactly

$$u - \theta D_1^2 w^2, \qquad u + (a_1 - \theta) D_1^2 w^2 \qquad (\theta \in \Theta).$$

The zero contact in the first list makes u one selected prime. The finite avoidance data include all denominators and contact collisions. The prescribed conjugate reduction orders and the compatible common-index progression make H, U formal at the primary places, and $H - P, V$ formal at the opposite places. Both full ratios equal n, so the reciprocal tests have depth at

least two and C, I have depth at least four. With $v(q_*) = -1$, every nonconstant factor term is positive at the selected place. At the opposite place each factor is either a unit of residue one or has negative valuation. Thus $\mathrm{Tr}(W^{-1})$ reduces to one or two and is a unit. The actual pole input has valuation one there.

Put $v_0 = D_1^2 w^2$. For $a \in \{0, a_1\}$ the exact identity is

$$R_*(\lambda + a) = \frac{\prod_{\theta \in \Theta}(u + (a - \theta)v_0)}{v_0^8(\lambda + a - 1)^{10}}.$$

All factors other than the selected prime numerators are rational squares. The trace changes every valuation by an even number. Consequently the pole input has square reduced denominator. At $2, 3, 17$ choose denominator divisibility so the argument dominates the fixed a_1 p-adically. Then $q_*, R_* \sim t^{-2}$, all weighted tests are small, and $\mathrm{Tr}(W^{-1})$ is arbitrarily close to two. This gives the required local squareclasses and depths in (1). On the compact real neighborhoods, choosing an arbitrarily large fixed lower bound on λ makes both arguments large and positive; $W \to 1$ and $b_0 \sim 1/(4t^2)$, so $0 < b_0 < 1$. These bounds are chosen before the prime selection. The application of Jaswal's simultaneous-twist theorem in [2] now supplies independent prepared pole witnesses for both blocks; the extra preparation at 3 and 17 is harmless.

No parity restriction on n was used: multiplication by an even or odd integer preserves the formal group, and the full ratios are the same integer n. The construction works for $X = 0$ as well. It proves completeness for every integer input and every allowed fixed shift. Together with the uniform soundness threshold it proves Theorem 1.1.

6 Counts, coefficients and verification

The retained rational-coordinate census is

Input, real comparison and slope	3
Common point coordinates f_H, S, T_H	6
Reciprocal pair and secant parameter	10
Two field ratios	4
Four reciprocal squares and two comparison squares	12
Trace and two height outputs	3
Two contact/pole blocks	40
Second argument	1
Total	79

The common squares occur in the fifth row of this census and are not counted twice. There are 71 rational quadratic equations. Clear each to a primitive integer row with positive first coefficient in the specified monomial order; their sum of squares is P_B.

The directory `computation/quartic/abc79_v27` contains every coordinate, every quadratic row and the expanded quartic. Collection gives 1140 parameter-separated terms and 1139 coordinate monomials for every positive B. The coefficient of X^4 is one, proving exact degree four and content one. The only mixed parameter profile is the constant coefficient $B^2 + C_0$; the largest fixed coefficient is C_*. All remaining parameter profiles are nonzero linear multiples of B and are bounded by their maximum. Exact endpoint and monotonicity inequalities in the independent checker prove the height formula for every $B \geq 3$.

The checker reconstructs the unchanged rows against the complete 81-variable source and the changed pole and reference rows from the exact field formulas. It independently expands all coefficients. The reference checker verifies five explicit ideal-membership certificates,

two converse identities, all six original source components, and the exceptional coefficient calculation; omitting the corrected $f_H S$ term is rejected by a negative control. The 702 exact point-pair examples retained from [4] include identity, tangent, torsion and half-integral cases. These are supplementary finite checks, not replacements for the universal proofs.

For the lower-coefficient 81-variable alternative, retain the complete older reference chart and change only the three nonsquare-definition pole rows in each block. The two charts have the same interpreted reference points with rational reconstruction in both directions. All other field values and pole equations agree. Thus the two new families have exactly the same rational input projection for every fixed B, even before the sufficiently large shift is selected. This does not assert projection equivalence with the old pole system. The full low-coefficient tables and checker are retained separately.

References

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[4] Companion paper, *Trace heights on quadratic twists with $j = 1728$*, 13 November 2026, `papers/trace_heights/trace_on_all_points_20261113_v25.tex`.

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